Nirvek Pandey

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Education

University of California San Diego **La Jolla, CA**
B.S. in Computer Science, *Regents' Scholar* September 2022 – December 2026

- **AI/ML:** Search & Learning Algorithms, Data Modeling, DL, NLP, Reccomender Systems and Web Mining
- **Systems:** OS, Networked Services, Parallel Computing, Graduate OS, Data Center Systems, Wireless Systems

Experience

Android Multimedia - Software Engineering Intern **Irvine, CA**
Panasonic Avionics Corporation June – September 2026

- Implemented streaming RTP over multicast support for AndroidX ExoPlayer deploying to over 2,500 commercial planes; optimized stream delay by 63% from 24 to 5 fps compared to legacy player.
- Created app for testing concurrent media playback, extended Python3 catalog API for SFVC and RTP cameras, and resolved multi-session state discontinuity between MediaService and Zenoh Pub/Sub control plane.
- Optimized fast-traversal of short form video content (reels) leveraging concurrent sessions, integrated custom MiniPlayer with MediaBrowser + PlayerViews, and reduced library loading by 85% through catalog caching.
- Deployed five C/C++ services (including advertising video on demand, Widevine DRM licensing, & RTSP daemon) integrating Conan artifactory, verifying with k3s, packaging with Gitlab pipelines on Converix VTE.

Computer Vision - Data Science Intern **Costa Mesa, CA**
Newton(Neurolens) June – September 2024

- Collaborated with data architects to implement an ETL system using SQL and Azure Pipelines, separating analytics workloads that improved pipeline reliability by 27% and accelerated R&D analysis.
- Built an ML pipeline with OpenCV and PyTorch to detect suppressed eye-tracking signals with 84% accuracy in identifying suppressed measurement and enabling more reliable data interpretation for optometry research.

Software Engineering Intern **La Jolla, CA**
UC San Diego April – June 2024

- Led a cross-functional Agile Scrum team of 8, facilitating weekly stand-ups and retrospectives that enhanced communication and reduced update turnaround times 46%, managing task progress using GitHub Projects.
- Designed CI/CD workflows with GitHub Actions, reducing deployment errors by 24% and accelerating releases through automated JUnit unit tests and Jest/Puppeteer-based integration and end-to-end testing.

Projects

Home Network Lab *Proxmox, Docker, Cloudflare, Bare Metal Hypervisor, Tailscale* **May 2026 – Present**

- Recycled gaming PC into Proxmox VMM for LXC/VM server hosting **personal portfolio**, Minecraft server, network wide DNS sinkhole, nginx server with HTTP message client for ESP32 and Pi based IoT projects.
- Developing C++ battleship simulator for custom dual-screen ESP32-C3 hardware and WASM emulator, leveraging Go based HTTP server for cross-device online gameplay and ESPnow for offline play.

Peer-to-Peer Coding Collaboration for VS Code *Go, Typescript, Tailscale* **April – June 2026**

- Built distributed system for preserving edit order under concurrent writes, achieving sub-110ms edit latency
- Handled leader failures with RAFT election, restoring service within 1-2s of leader failure
- Workspace and edit history addressed within 2s of user reconnecting, updating from write-ahead log.

Research - OS Performance Analysis *Linux, C, ARM64, Python* **September – December 2025**

- Developed low-level benchmarking suite in C for Raspberry Pi 5 (ARM architecture) testing CPU scheduling, memory hierarchy, TCP/IP protocols, and file system I/O performance across multiple operating system layers.
- Quantified storage bottlenecks by measuring a 500x latency spike for disk-based page faults and identified a 6.4 GB file buffer cache limit using pointer chasing to map the L1/L2/L3 cache and DRAM hierarchy.
- Benchmarked network performance using ping, ethtool, and custom socket programs, discovering an 8x latency penalty for remote file access over WiFi compared to local disk operations.

TritonTube *Go, SQLite, gRPC, etcd, AWS EC2, FFmpeg, MPEG-DASH, HTTP* **May – July 2025**

- Engineered fault-tolerant video platform with RESTful APIs for upload/playback, implementing MPEG-DASH adaptive streaming to optimize video quality and reduce buffering by over 50% across varying network conditions.
- Built scalable metadata and content services on AWS EC2 with etcd-backed RAFT consensus protocol, ensuring data consistency and high availability while supporting 100+ concurrent users for video processing workloads.

Skills

- **Programming Languages:** Kotlin, Java, Python, Go, JavaScript/TypeScript, C, C++, SQL, HTML, CSS
- **Libraries & Frameworks:** PyTorch, NumPy, Pandas, Matplotlib, TensorFlow, Scikit-Learn, OpenCV, React, Next.js, Angular, Express.js, Node.js, Flask, Tailwind CSS, JUnit, Jest, Puppeteer, FFmpeg, gradle, Conan
- **Tools & Technologies:** Linux, Wireshark, Git, Docker, gRPC, etcd, Kubernetes, Helm, VLC, PostgreSQL, Firebase, Redis, GCP, Google Cloud Run, Android SDK, adb, LeakCanary, Media3, ExoPlayer, Jetpack Compose, logcat, Vercel, Gitlab Pipelines, AWS EC2, AWS S3, WandB, CUDA, OpenCL, MPEG-DASH, MPEG2-TS, HLS, RTP, RTSP, GitHub Actions, SharePoint, Excel, PowerPoint, BLE, WiFi, LoRA, Thread, Zigbee